

Day 56 - Wireless Architectures

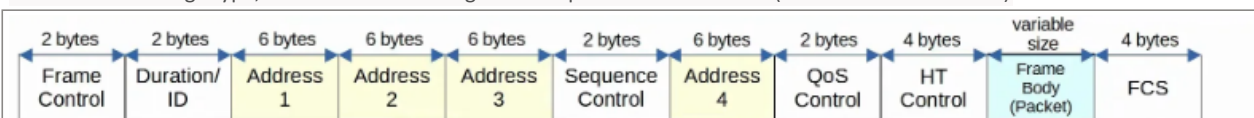
Wednesday, September 9, 2026

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Wireless Architectures Info

802.11 - Frame Format

802.11 frames have a different format than the 802.3 Ethernet frames. The frame can change depending on the 802.11 version and message type, some of the fields might not be present in the frame. (like the 4 address fields)



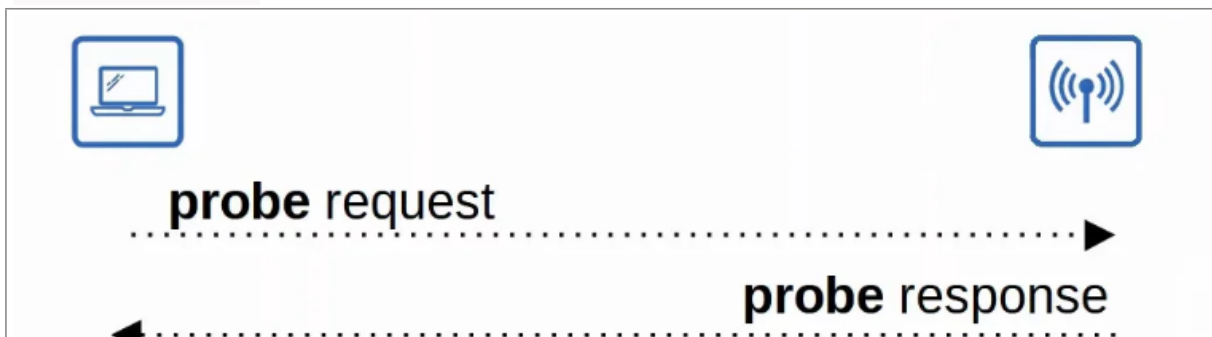
802.11 Frame Format

- 1. Frame Control** - (2 bytes)
 - Provides information such as the *message type* and *subtype*
- 2. Duration/ID** - (2 bytes)
 - Depending on the *message type*, this field can indicate:
 - Duration*, the time (in microseconds) the channel will be dedicated for transmission of the frame.
 - ID*, the identifier for the association (connection).
- 3. Address 1/2/3/4** - (6 bytes)
 - Up to *four* addresses can be present in an **802.11** frame. It depends on the *message type*.
 - Destination Address (DA)*: Final recipient of the frame
 - Source Address (DA)*: Original sender of the frame
 - Receiver Address (DA)*: Immediate recipient of the frame
 - Transmitter Address (DA)*: Immediate sender of the frame
- 4. Sequence Control** - (2 bytes)
 - Used to reassemble fragments and eliminate duplicate frames.
- 5. QoS Control** - (2 bytes)
 - Used in **QoS** to prioritize certain traffic.
- 6. HT Control** - (4 bytes)
 - Added in **802.11n** to enable *High Throughput* operations.
 - 802.11n** is also known as '*High Throughput*' (*HT*) Wi-Fi.
 - 802.11ac** is also known as '*Very High Throughput*' (*VHT*) Wi-Fi.
- 7. Frame Body (Packet)** - (*variable size in bytes*)
 - Contains the actual packet being sent over the network.
- 8. FCS** - (4 bytes)
 - Used to check for errors, same as in the Ethernet frame.

802.11 - Association Process

APs are used to bridge traffic between wireless stations and other devices.

- For a station to send traffic through the **AP**, it must be *authenticated* and *associated* with the **AP**.
- There are three **802.11** connection states:
 - Not *authenticated*, not *associated*
 - Authenticated*, not *associated*
 - Authenticated*, *associated*





Station scanning modes

- There are two ways a station can scan for a **BSS**:
 1. **Active Scanning**
 - The station sends *probe requests* and listens for a *probe response* from an **AP**.
 2. **Passive Scanning**
 - The station listens for *beacon* messages from an **AP**.
 - *Beacon* messages are sent periodically by **APs** to advertise the **BSS**.

802.11 - Message Types

There are three 802.11 message types:

1. **Management**
 - Used to manage the **BSS**.
 - Includes the following messages:
 - *Beacon*
 - *Probe request, probe response*
 - *Authentication request, authentication response*
 - *Association request, association response*
2. **Control**
 - Used to control access to the medium (radio frequency).
 - Assists with the delivery of **management** and **data** frames.
 - Includes the following messages:
 - *Request To Send (RTS)*
 - *Call To Send (CTS)*
 - *ACK*
3. **Data**
 - Used to send actual data packets.

Wireless Access Point (AP) Deployments

There are three main wireless AP deployment methods:

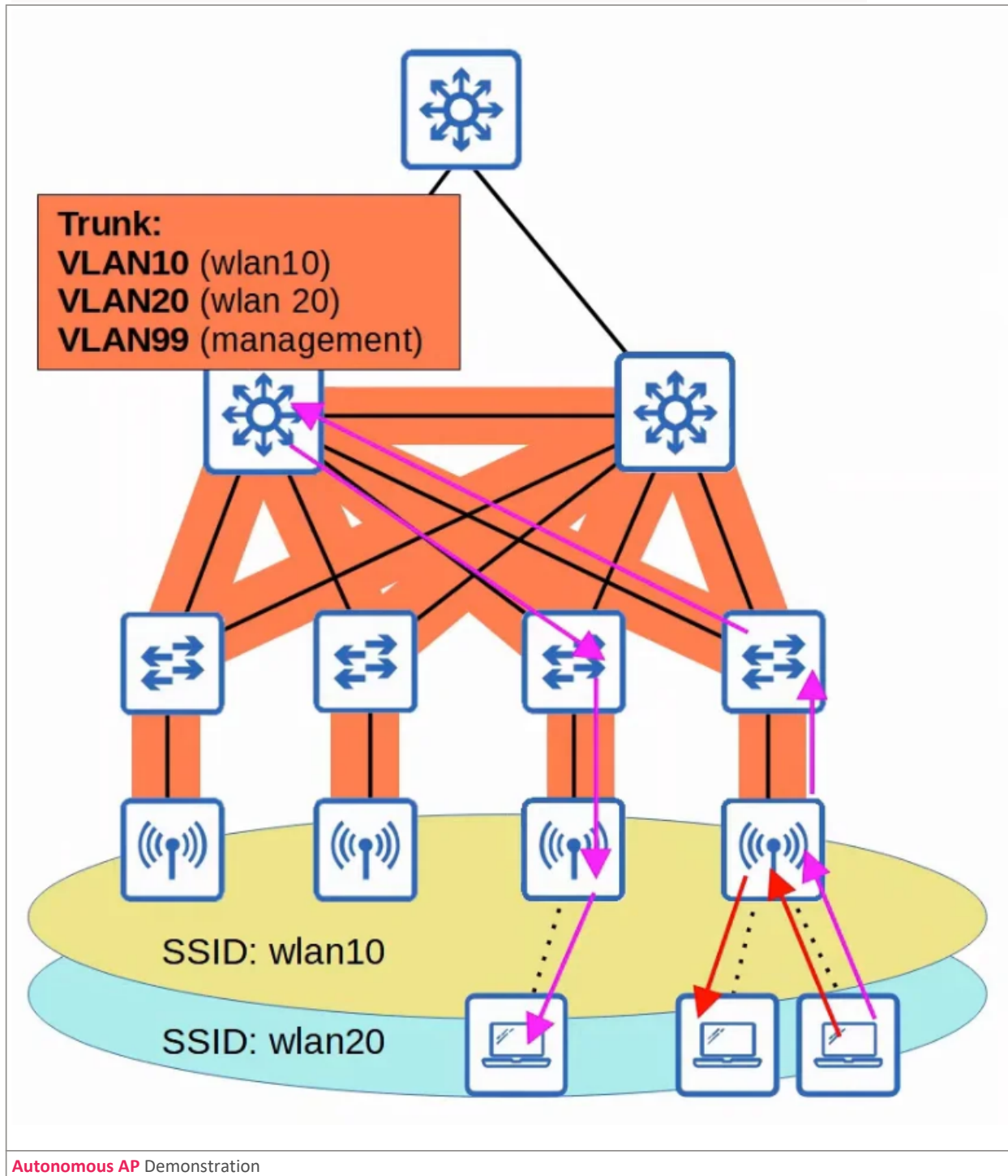
1. **Autonomous**
2. **Lightweight**
3. **Cloud-based**

1. Autonomous

Autonomous APs are self-contained systems that don't rely on a Wireless LAN Controller (WLC).

- They are configured individually, and there is no central monitoring or management of **APs**.
 - Can be configured through console cable (CLI), **SSH/Telnet** (CLI), or **HTTP/HTTPS** web connection (GUI).
 - An IP address for remote management should be configured.
 - Radio frequency parameters must be manually configured.
 - Security policies are configured and handled individually by each **AP**.
 - **QoS** rules and etc. are configured and handled individually by each **AP**.
- **Autonomous APs** connect to the *wired* network with a *trunk* link
- Data traffic from the *wireless* clients will have a direct path to the *wired* network or to any other *wireless* clients associated with the same **AP**.
- Each VLAN has to stretch across the entire network. This is considered bad as it leads to a few issues:
 - Large broadcast domains.

- **STP** will disable links.
- Adding/deleting VLANs is very labor-intensive
- **Autonomous APs** can be used for small networks, but they are not very viable in medium to large networks because of scalability issues.
- **Autonomous APs** can function in any of the **wireless AP** modes (**Repeater**, **outdoor bridge**, **workgroup bridge**).



2. Lightweight

In this deployment mode, the functions of an AP can be split between the AP and a Wireless LAN Controller (WLC).

- **Lightweight APs** handle 'real-time' operations (transmitting/receiving RF traffic, encryption/decryption, sending out beacon/probe messages, etc.)
- Other functions are carried out by the **WLC** (RF management, security/**QoS** management, client authentication/association, roaming management, etc.)
- This is called **Split MAC Architecture**, as the operations are split between the **AP** and the **WLC**.

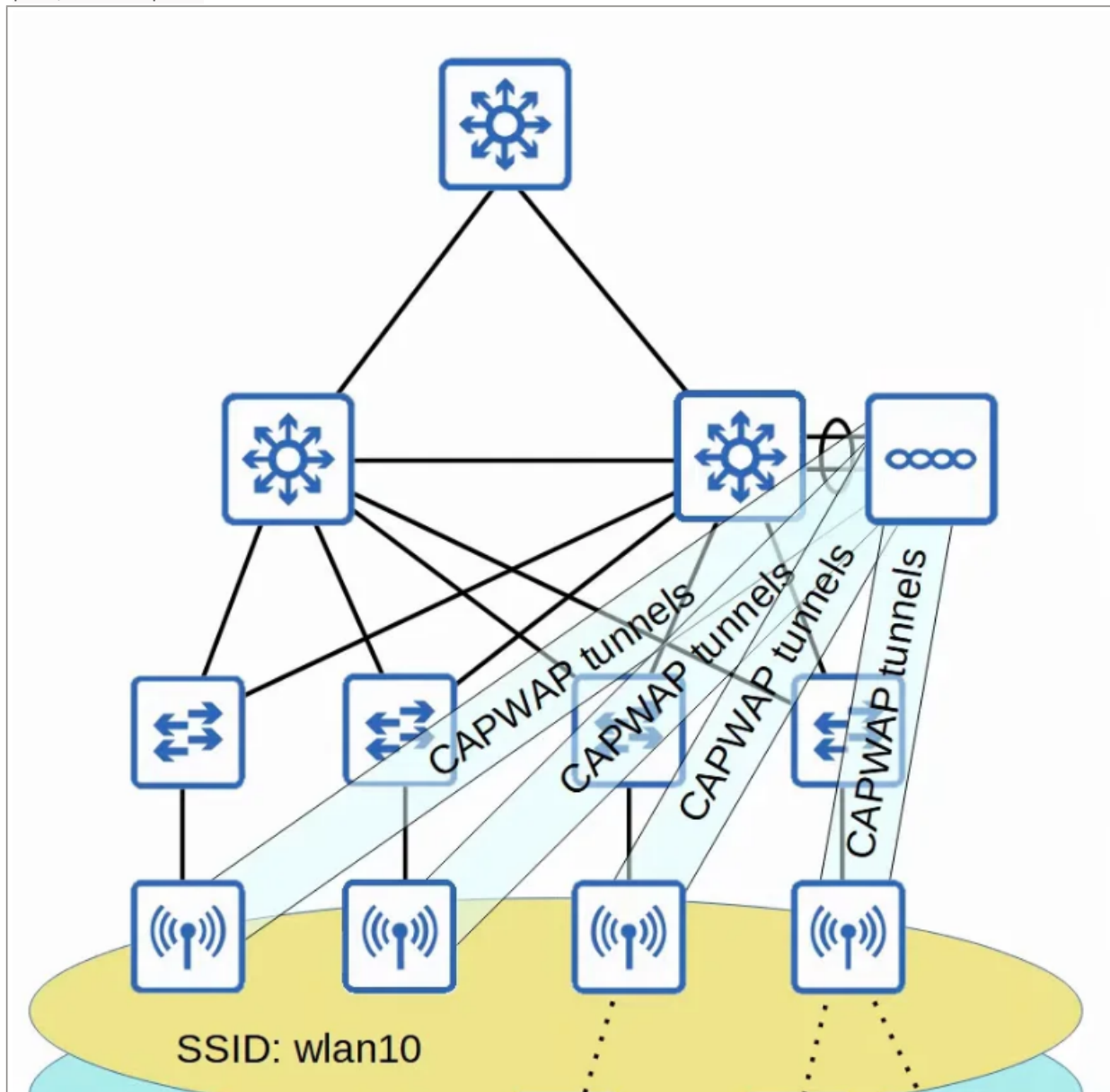
- This is called **Split-MAC Architecture**, as the operations are split between the **AP** and the **WLC**.

WLC Information

- The **WLC** is also used to centrally configure the **lightweight APs**.
- The **WLC** can be located in the same subnet/VLAN as the **lightweight APs** it manages, or in a different subnet/VLAN.
- The **WLC** and **lightweight APs** authenticate each other using **digital certificates** installed on each device (X.509 standard certificates).

Lightweight AP Information

- The **WLC** and **lightweight AP** use a protocol called **Control and Provisioning of Wireless Access Points (CAPWAP)** to communicate with each other.
 - Based on an older protocol called **Lightweight Access Point Protocol (LWAPP)**.
- Two **CAPWAP** tunnels are created between each **AP** and the **WLC**:
 1. **Control** tunnel (UDP port **5246**):
 - This tunnel is used to configure the **APs**, and control/manage the operations.
 - All traffic in this tunnel is **encrypted** by default.
 2. **Data** tunnel (UDP port **5247**):
 - All traffic from wireless clients is sent through this tunnel to the **WLC**. *(It does not go directly to the wired network)*
 - Traffic in this tunnel is not **encrypted** by default, but you can configure it to be **encrypted** with **Datagram Transport Layer Security (DTLS)**.
- Because all traffic from **wireless** clients is tunneled to the **WLC** with **CAPWAP**, **APs** connect to switch ==**access**== ports, not **trunk** ports.



SSID: wlan20



Lightweight AP Demonstration

Lightweight Operation Modes

Lightweight APs can be configured to operate in various modes:

1. Local

- This is the default mode where the AP offers a BSS (more multiple BSSs) for clients to associate with.

2. FlexConnect

- Like a lightweight AP in Local mode, it offers one or more BSSs for clients to associate with. However, FlexConnect allows the AP to locally switch traffic between the wired and wireless networks if the tunnels to the WLC go down.

3. Sniffer

- The AP does not offer a BSS for clients.
- It is dedicated to capturing 802.11 frames and sending them to a device running software such as Wireshark.

4. Monitor

- The AP does not offer a BSS for clients. It is dedicated to receiving 802.11 frames to detect rogue devices.
- If a client is found to be a rogue device, an AP can send de-authentication messages to disassociate the rogue device from the AP.

5. Rogue Detector

- The AP does not even use its radio. It listens to traffic on the wired network only, but it receives a list of suspected rogue clients and AP MAC addresses from the WLC.
- By listening to ARP messages on the wired network and correlating it with the information it receives from the WLC, it can detect rogue devices.

6. SE-Connect (Spectrum Expert Connect)

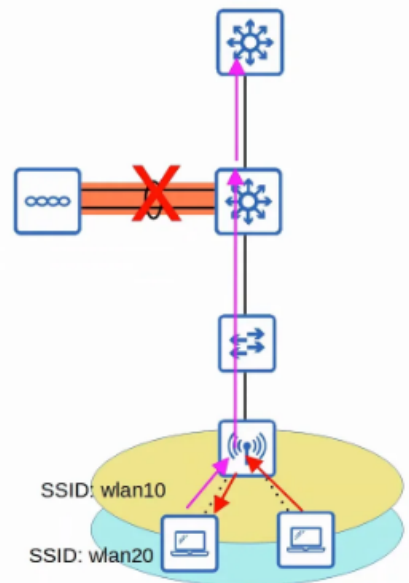
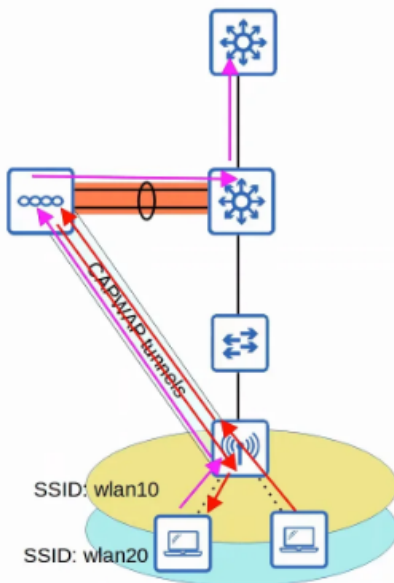
- The AP does not offer a BSS for clients.
- It is dedicated to RF spectrum analysis on all channels. It can send information to software such as Cisco Spectrum Expert on a PC to collect and analyze the data.

7. Bridge/Mesh

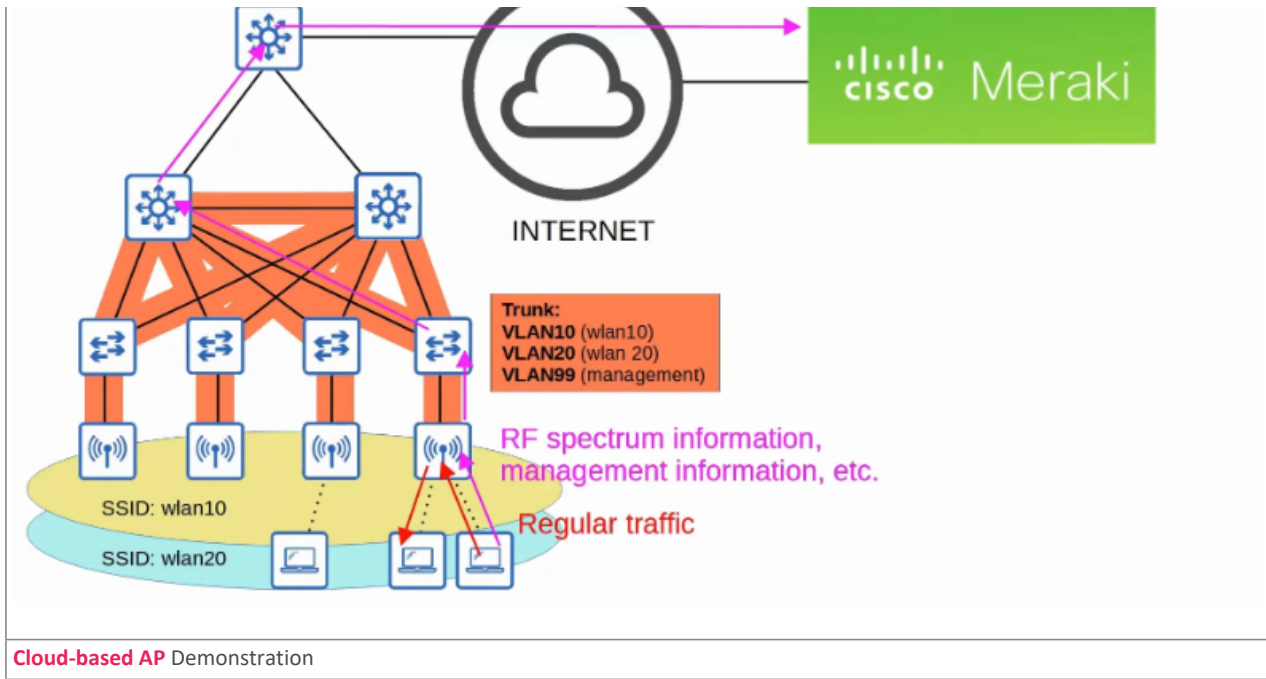
- Like the autonomous AP's Outdoor Bridge, the lightweight AP can be a dedicated bridge between sites, for example over long distances.
- A mesh can be made between the APs.

8. Flex plus Bridge

- Adds FlexConnect functionality to the Bridge/Mesh mode.
- Allows wireless APs to locally forward traffic even if connectivity to the WLC is lost.



Local / FlexConnect



Wireless LAN Controller (WLC) Deployments

In a Split-MAC Architecture (Lightweight AP deployment), there are four main WLC deployment models:

1. **Unified**
 - Supports up to **6000 APs**.
 - The **WLC** is a hardware appliance in a central location of the network.
2. **Cloud-based**
 - Supports up to **3000 APs**.
 - The **WLC** is a **VM** running on a server, usually in a private cloud in a data center.
 - (This is not related to the [Wireless AP Cloud-based deployment](#))
3. **Embedded**
 - Supports up to **200 APs**.
 - The **WLC** is integrated within a switch.
4. **Mobility Express**
 - Supports up to **100 APs**.
 - The **WLC** is integrated within an **AP**.

Extra Tip: WLC and EtherChannels

- **WLCs** only support **static LAG (EtherChannel)**.
- Therefore, they can't run **PAgP** or **LACP**.